Nucleic Acids Research, 2026, pp. 1–8

doi: DOI HERE
Advance Access Publication Date: Day Month Year
Computational Biology

COMPUTATIONAL BIOLOGY

Benchmarking the accuracy, robustness and reproducibility of spatial domain identification methods

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FOR PUBLISHER ONLY Received on Date Month Year; revised on Date Month Year; accepted on Date Month Year

Abstract

Spatial domain identification methods are usually compared by small differences in adjusted Rand index (ARI), mostly on one tissue and from a single run. We benchmarked 13 published methods and three training-free baselines on 20 annotated sections from 10x Visium, STARmap and MERFISH, and separated each learned embedding from its clustering step. When every embedding was clustered in the same way, a training-free neighbourhood-smoothing baseline had a higher mean ARI than 11 of 16 comparators and could not be separated from the other 5, which included BASS and BANKSY; no comparator was equivalent to it. The clustering step itself was a large source of variation: on fixed embeddings, refitting the same Gaussian mixture from different random starts changed ARI by 0.18 on average (up to 0.60), and the default fit of R mclust, used by several methods, depended on the random seed through its subsampled initialisation (up to 0.20). Keeping the most likely of 20 fits helped on human cortex but not on single-cell MERFISH data, where likelihood and ARI were unrelated. HiSTaR did not reproduce its published accuracy with its official code, and the released code of one method selects its reported epoch using the labels. We recommend reporting the clustering protocol and its variance, never selecting runs with labels, and including training-free baselines.

Key words: spatial transcriptomics, spatial domains, clustering, reproducibility, benchmarking

Introduction

Spatial transcriptomics measures gene expression at thousands of locations in a tissue section while retaining their coordinates. Identifying *spatial domains*, regions with coherent expression such as cortical layers, is one of the first analyses of such data, and many methods have been proposed for it, from Bayesian models [1] and neighbourhood-augmented features [2] to graph neural networks [3; 4; 5] and hierarchical graph variational autoencoders such as HiSTaR [6].

Most of these methods are evaluated on the twelve manually annotated human dorsolateral prefrontal cortex (DLPFC) sections [7] and report median adjusted Rand index (ARI) [8] improvements of a few hundredths over previous methods [4; 5; 6]. Several (for example GraphST, STAGATE and SEDR) share the same last step: the learned embedding is clustered once with the EEE Gaussian mixture of the R package mclust [9]. Benchmarks have found that rankings depend strongly on the dataset [10; 11], that deep methods vary with the random seed [12], that replacing each method's clustering algorithm changes results [11], and that preprocessing and clustering choices often matter more than architectural novelty [13]. Simple training-free pipelines can also be competitive [14; 15]. What these studies do not measure directly is how much of the variation comes from the clustering step itself when the embedding is held fixed.

We benchmark 13 published methods and three training-free baselines along three axes. *Accuracy*: how do the methods compare across three technologies when their embeddings are clustered in the same way, and how does this change with each method's own clustering? *Robustness*: on a *fixed* embedding, how much does ARI change with the random initialisation of the mixture model and with the random seed of R mclust, and does selecting fits by likelihood help? *Reproducibility*: do published results reproduce with the official code (here HiSTaR's DLPFC results), and does released code select models using the labels? The study design is summarised in Figure 1. We find that a standard neighbourhood-smoothing baseline is among the most accurate methods, that the clustering step varies more than most differences between methods, and that released code can use the labels for model selection.

Materials and Methods

Data

DLPFC (10x Visium). Twelve sections from three donors [7]; seven annotated layers (five for donor 2),

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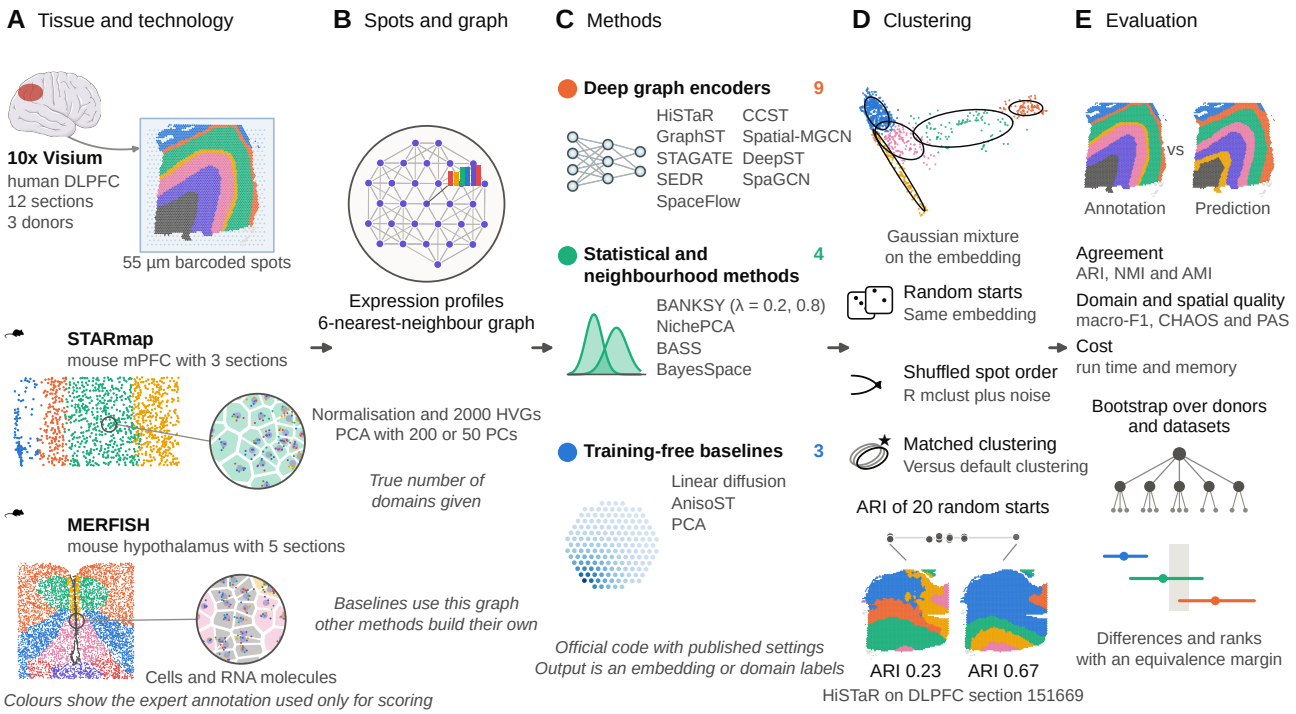


Fig. 1. Study design. (**A**) Human DLPFC on a 10x Visium capture area (section 151673) and mouse sections profiled by STARmap (mPFC, section BZ5) and MERFISH (hypothalamus, Bregma −0.14); colours are the expert annotation, used only for scoring. (**B**) Each spot or cell carries an expression profile; the training-free baselines use the six-nearest-neighbour graph, and other methods build their own graph as documented. The true number of domains is given to every method. (**C**) 13 published methods and three training-free baselines in three families (family colours are used in Figures 1, 2 and 5). (**D**) Each embedding is clustered with the method's default clustering and with a matched clustering (the same 20-start Gaussian mixture for every method); single-start refits and R mclust with different random seeds measure the variance of the clustering step. Below: one fixed HiSTaR embedding of DLPFC section 151669 refitted from 20 random starts (ARI of each start) and the domains of the worst and best start. (**E**) Predicted domains are compared with the annotation using agreement, per-domain, spatial and cost metrics, summarised with a bootstrap over donors and datasets. Icons: Bioicons (CC0) and Servier Medical Art (CC BY 3.0).

`layer_guess_reordered`. **STARmap mouse medial prefrontal cortex.** Three sections (about 1050 cells, 166 genes) with four annotated layers (L1, L2/3, L5, L6) [16]. **MERFISH mouse hypothalamus.** Animal 1, five sections from Bregma −0.04 to −0.24 (about 5500 cells, 155 genes) with eight annotated regions [17]. The STARmap and MERFISH annotations are those distributed with the BASS analyses [18], one of the benchmarked methods (see Limitations). In total, 20 sections from three technologies (Supplementary Table S1; Supplementary Note 1).

Preprocessing

DLPFC: genes in fewer than 50 spots or with fewer than 10 counts were removed, 2000 highly variable genes were selected (Seurat v3 flavour), counts were library-size normalised, log-transformed and scaled, and reduced to 200 principal components (PCs). STARmap and MERFISH (targeted panels): all genes, same normalisation, 50 PCs. All methods received the same input where they accept one; BANKSY used the log-normalised genes, and SpaGCN, GraphST and the other published methods used their own documented preprocessing of the same raw counts. The true number of domains was given to every method, and unannotated spots were excluded from evaluation.

Methods

Methods were chosen to cover the main families of spatial domain identification with maintained public code that runs on CPU: deep graph encoders, statistical and neighbourhood methods, and Bayesian models. Every method was run with its official code and default or tutorial settings on one CPU thread; none was tuned per dataset. Versions, settings and deviations from the tutorials (for example the neighbourhood radius, expressed in spot spacings so that it transfers across technologies) are listed in Supplementary Table S2.

- **Deep graph encoders.** HiSTaR [6] (PyTorch port of the official code with the published defaults: cross-level weight 0.3, 200+200 epochs), GraphST [5] (official package and tutorial: 600 epochs, 20-PC projection, R mclust; with and without its label refinement), SpaGCN [3] (tutorial without histology, Louvain resolution search to the true number of domains), STAGATE [4], SEDR [19], SpaceFlow [20], CCST [21], Spatial-MGCN [22] and DeepST [23]. Spatial-MGCN's released test script reports the epoch with the highest ARI against the annotation; in the benchmark we report its final epoch (label-free) and give the label-selected value separately.
- **Statistical and neighbourhood methods.** BANKSY [2] (official `pybanksy`, 18 neighbours, $m_{\max} = 1$, 20 PCs; $\lambda = 0.8$, recommended for domain segmentation, and $\lambda = 0.2$, recommended for cell typing, included

for reference), NichePCA [14] (a published training-free method: neighbour aggregation and PCA), and the Bayesian models BASS [18] and BayesSpace [1], which cluster internally. BayesSpace requires a spot lattice and was run on the Visium sections only.

- **Training-free baselines.** *Linear diffusion* is standard neighbourhood smoothing (graph Laplacian smoothing): the top 30 PCs are smoothed on the spatial six-nearest-neighbour graph, $H^{(t+1)} = (1-\alpha)H^{(t)} + \alpha D^{-1}AH^{(t)}$, $\alpha = 0.5$, $T = 10$. *AnisoST* uses the same scheme with edge weights recomputed at every step, $w_{ij}^{(t)} = \exp(-\|h_i^{(t)} - h_j^{(t)}\|^2/\kappa^{(t)})$ (Perona–Malik-type edge-preserving diffusion [24]; $\kappa^{(t)}$ the median squared edge difference); a similar idea has been used inside a deep model [25]. *PCA* without smoothing is the non-spatial reference. All baseline settings were fixed on DLPFC donor 1 before any other data were analysed (sensitivity: Supplementary Fig. S2).

Seeds: three per section for most methods; one for STAGATE, CCST, Spatial-MGCN and BASS in the published pipelines, and one run per section for BayesSpace (Supplementary Table S19).

Neutrality.
The authors developed AnisoST, and linear diffusion and PCA are its ablations; the authors did not develop any of the published methods. To limit bias, all baseline settings were frozen on four development sections, every published method was run with its authors' defaults, and all comparisons are repeated without the development sections.

Clustering protocols

Default clustering: the clustering step of each method's official pipeline. This is R mclust (EEE) for GraphST, STAGATE and SEDR; a 3-start scikit-learn Gaussian mixture with tied covariance [26] for HiSTaR, BANKSY and the three baselines; Leiden or k-means for SpaceFlow, NichePCA, CCST, Spatial-MGCN and DeepST; and the internal model for SpaGCN, BASS and BayesSpace. *Matched clustering* (primary analysis): every embedding clustered with the same tied-covariance Gaussian mixture (the mclust EEE model) from 20 random starts, keeping the fit with the highest log-likelihood (label-free). SpaGCN, BASS and BayesSpace output no embedding and keep their own clustering under both protocols. *Initialisation experiment*: one seed-0 embedding per section and method, refitted 20 times with a single random start each (14 methods and baselines that output an embedding). *R mclust experiment*: R mclust 6.0.1 (`Mclust(Z, G = k, modelNames = "EEE")`) on the same embeddings, (i) with its default initialisation under ten R random seeds and fixed row order, (ii) initialised on all observations (`initialization = list(subset = 1:n)`), and (iii) the same after adding Gaussian noise with 1% of each feature's standard deviation, five times.

Evaluation and statistics

ARI against the manual annotation is the primary metric; NMI gave the same ranking of methods (Supplementary Table S4), and further agreement, spatial and label-free metrics are in Supplementary Table S12 (definitions in Supplementary Note 3). For every method and section we average over seeds. Sections are not independent (the 12 DLPFC sections come from three donors; the STARmap and MERFISH sections come from one experiment each), so the unit of analysis is the group: 5 groups (three DLPFC donors, STARmap, MERFISH). For each method we estimate the difference in mean ARI from linear diffusion, the reference, which was chosen after the means were seen, so these comparisons are exploratory. Intervals come from a hierarchical bootstrap (5000 resamples of groups, then of sections within a group; the estimate is the equal-weight mean of group means) with 95% percentile intervals (Supplementary Note 4). A method is called equivalent to the reference if the 90% interval lies within ±0.03 ARI, about the seed-to-seed standard deviation. Ranks and the probability of ranking first come from the same bootstrap. The analysis is repeated without the development sections (DLPFC donor 1 and the first section of STARmap and MERFISH). With five groups the intervals are wide and no correction for multiple comparisons is applied; we therefore interpret only consistent patterns. Run time and peak memory were measured in a standardised runner that gives every method the same input (process-tree sampling, one CPU thread per job; Supplementary Note 6); its accuracy agrees with the published pipelines to within the differences listed in Supplementary Table S14.

Results

A training-free smoothing baseline is among the most accurate methods

With matched clustering, linear diffusion had a mean ARI of 0.558, 0.705 and 0.497 on DLPFC, STARmap and MERFISH (Figure 2A; per-dataset values for both protocols in Supplementary Table S3). Across all methods and both protocols, the highest dataset means were reached by linear diffusion on DLPFC (0.558), BANKSY $\lambda = 0.8$ on STARmap (0.740) and BASS on MERFISH (0.582). Linear diffusion had a higher mean ARI than 11 of 16 comparators (95% interval above zero; Figure 2B; Supplementary Table S6), and this held for 11 of them without the development sections. Five comparisons were unresolved: BASS (Δ +0.025, 95% interval −0.044 to +0.101), BANKSY $\lambda = 0.8$ (+0.044, −0.007 to +0.095), HiSTaR (+0.064, −0.018 to +0.163), SEDR (+0.164, −0.007 to +0.352) and AnisoST (+0.048, −0.007 to +0.105). No comparator was equivalent to the baseline within ±0.03 ARI, and BASS was ahead on STARmap and MERFISH. Averaging the three dataset means with equal weight instead of the five groups puts BASS (0.594) slightly above linear diffusion (0.587). In the bootstrap, linear diffusion ranked first in 0.60 of the resamples (95% rank interval 1–3 among 16 methods) and BASS in 0.25 (Figure 2C; Supplementary Table S7). With each method's default clustering the picture was similar but more favourable to the baseline: 14 of 16 comparisons favoured it and 2 (BASS and BANKSY $\lambda = 0.8$) were unresolved (Supplementary Table S5). Example partitions are shown in Figure 3.

Seed-to-seed variability differed between methods: HiSTaR's standard deviation of ARI was 0.047 on DLPFC and 0.18 on STARmap, against 0.003–0.04 for the training-free baselines (Supplementary Table S8). The median time per section on one CPU thread ranged from under 30 s for the baselines and BANKSY to about 40 min for CCST (Supplementary Table S13). The edge-preserving variant AnisoST did not improve on linear diffusion overall (MERFISH: 0.363 vs 0.497), although on DLPFC it was more accurate on spots at layer boundaries (by 0.025, 9 of 12 sections, Wilcoxon signed-rank $p = 0.034$, uncorrected; Supplementary Fig. S3); at single-cell resolution neighbouring cells often differ by cell type, so

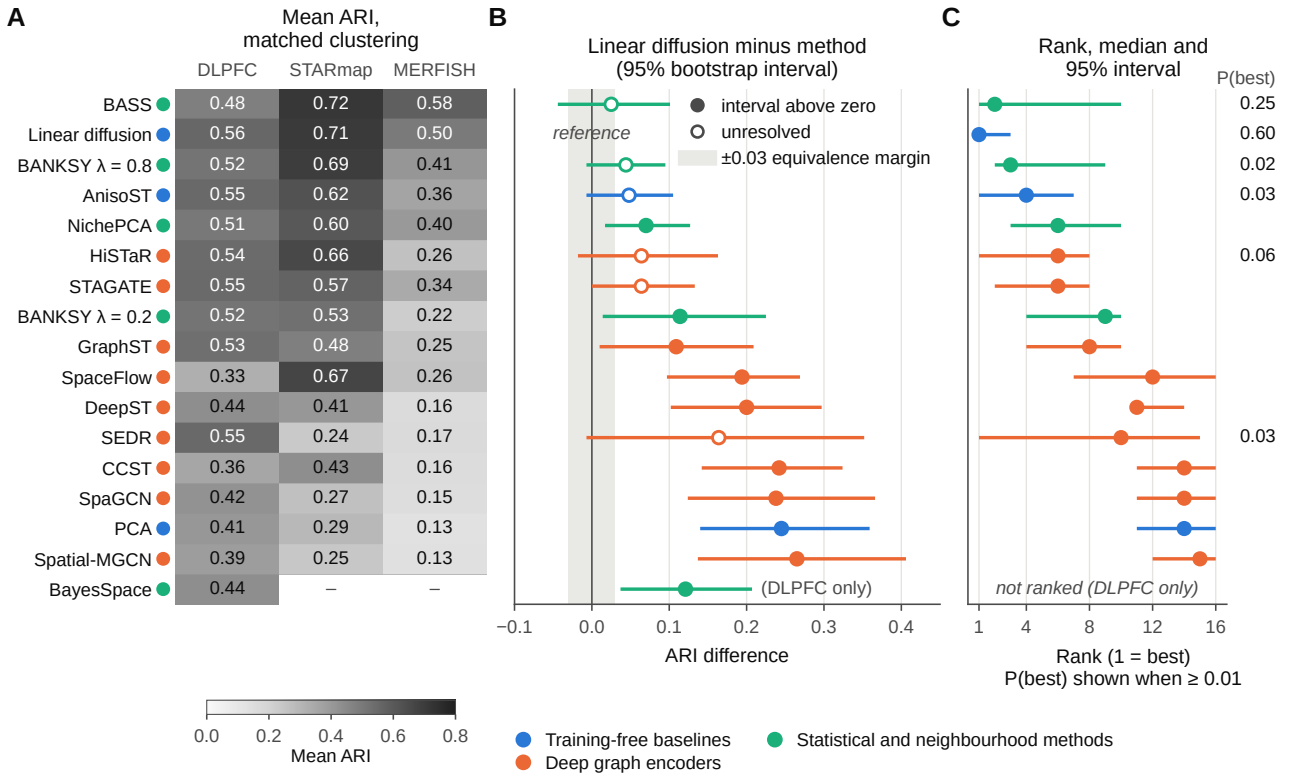


Fig. 2. Accuracy with matched clustering (every embedding clustered with the same 20-start Gaussian mixture; SpaGCN, BASS and BayesSpace use their internal clustering). $n = 20$ sections (12 DLPFC, 3 STARmap, 5 MERFISH); values are means over seeds. (**A**) Mean ARI per dataset; rows ordered by the mean over datasets; dots mark the method family. (**B**) Linear diffusion minus each method, with 95% intervals from a hierarchical bootstrap over 5 groups (three DLPFC donors, STARmap, MERFISH). Filled symbols: interval above zero; open symbols: unresolved. Grey band: equivalence margin of ±0.03 ARI. BayesSpace was run on DLPFC only. (**C**) Rank in the bootstrap (dot: median; line: 95% interval); numbers give the probability of ranking first when at least 0.01. Linear diffusion was chosen as the reference after seeing the means, so this comparison is exploratory.

an edge-preserving rule blocks the smoothing needed to reveal regions.

Random initialisation of the clustering step changes ARI on a fixed embedding

Figure 4A shows the ARI of 20 single-start fits of the same mixture model on each fixed embedding. Across the 280 embeddings of 14 methods and baselines, the ARI range within an embedding was 0.18 on average and reached 0.60 (SEDR, DLPFC section 151672). The training-free baselines were not immune: the mean range was 0.20 for linear diffusion, against 0.09 for SpaceFlow and 0.24 for BANKSY $\lambda = 0.8$ (Supplementary Table S9). The best of the 20 fits by ARI exceeded the average single fit by 0.08: choosing a clustering run by its agreement with the labels, explicitly or implicitly by trying several seeds, would inflate a result by more than the few hundredths typically reported between successive methods. Every fit for three representative methods is shown in Supplementary Fig. S1.

The default fit of R mclust depends on a hidden random seed

For data with more than 2000 observations, R mclust initialises the EM algorithm by hierarchical agglomeration on a random subset of 2000 observations drawn with R's random number generator, so its default fit depends on the R seed. The published pipelines fix this seed (2020 in STAGATE and GraphST, 2023 in SEDR), which makes their result reproducible but tied to an arbitrary choice. Across the 13 embeddings with more than 2000 spots or cells, changing only the R seed changed ARI by 0.11 on average (up to 0.20; Figure 4B), and the range exceeded 0.1 for 62% of them; the STARmap sections, with about 1050 cells, were initialised on all cells and did not vary. Initialising on all observations removed the dependence on the seed and on the row order; the remaining sensitivity to 1% noise was 0.06 on average (up to 0.12). The full-data fit fell outside the range of the ten seeded fits for 15% of the embeddings (Supplementary Table S10). An earlier version of this experiment attributed the variation to the order of the spots; permuting the rows changes which observations are sampled, so it is the same effect.

Selecting fits by likelihood helps on cortex but not on single-cell data

A natural fix is to run many starts and keep the most likely fit. On the same embedding and model, the best-likelihood fit beat the average single start for 10 of 14 methods and was worse for 3 (Supplementary Table S9). Replacing each method's default clustering by matched clustering changed mean ARI by more than 0.005 upwards for 8 of 14 methods and downwards for 5 (Figure 5; Supplementary Table S11); for example HiSTaR gained +0.053 ($p = 0.004$) and NichePCA +0.080 ($p = 0.040$), whereas SEDR changed by −0.092 ($p = 0.070$) and CCST

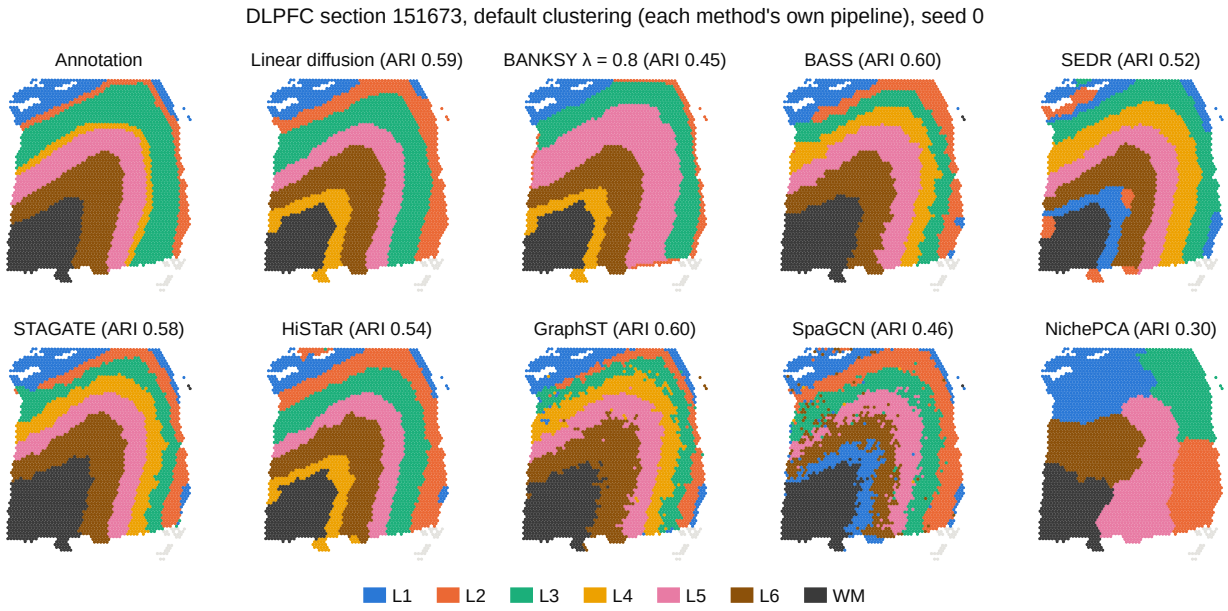


Fig. 3. Domains on DLPFC section 151673 (donor 3, not used for development; seed 0, default clustering). The manual annotation and the partitions of nine methods spanning the accuracy range; predicted clusters were matched to the annotated layers with the Hungarian algorithm for display only, and the ARI is given in each title. Spots without annotation are light grey.

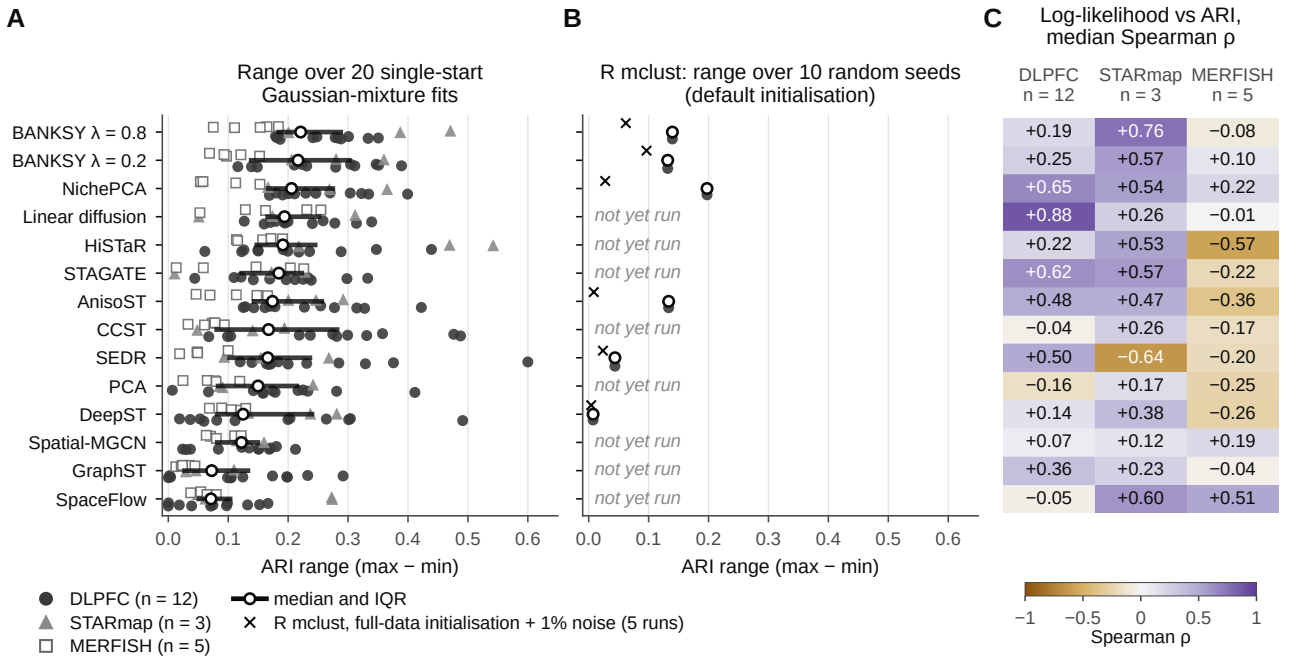


Fig. 4. Variance of the clustering step on fixed embeddings (14 methods and baselines that output an embedding; one symbol per section, $n = 20$ per method; shape and grey level: dataset). (**A**) ARI range over 20 single-start fits of the same Gaussian mixture. (**B**) ARI range of R mclust with its default initialisation over ten R random seeds; STARmap sections have fewer than 2000 cells, are initialised on all cells and show no variation. (**C**) Median Spearman correlation between the log-likelihood and the ARI of the 20 fits in A (positive: better-fitting mixtures agree better with the annotation); columns rest on 12, 3 and 5 sections. Bars: interquartile range; open circle: median.

by −0.071 ($p = 0.008$), all uncorrected. The reason is visible in Figure 4C: likelihood and ARI were positively related on DLPFC (median Spearman ρ 0.38) but not on MERFISH (median ρ −0.11; negative for 10 of 14 methods), where the best-fitting mixture can separate cell types rather than anatomical regions. Better optimisation of the mixture model is therefore not the same as better domains, and the clustering protocol alone can reorder methods.

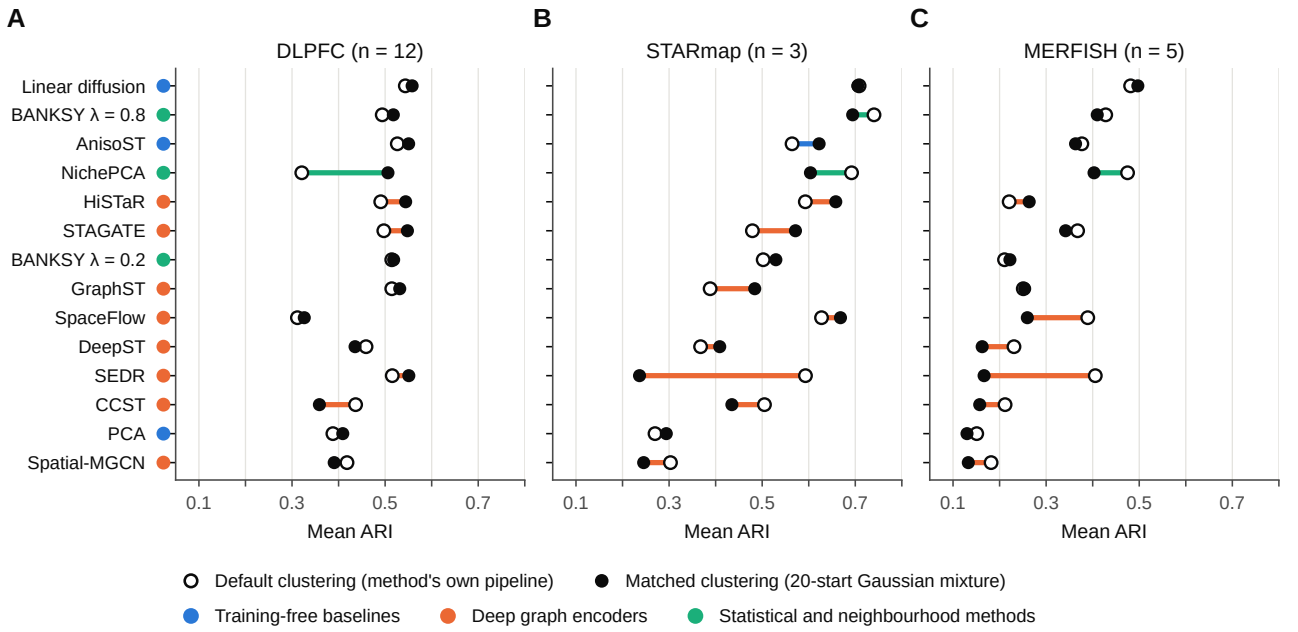


Fig. 5. Effect of the clustering protocol on mean ARI: (**A**) DLPFC, (**B**) STARmap, (**C**) MERFISH (means over 12, 3 and 5 sections). Open circles: default clustering; filled circles: matched clustering of the same embedding; line colour: method family. Rows are ordered by the matched mean. SpaGCN, BASS and BayesSpace cluster internally and are not shown.

HiSTaR's published DLPFC accuracy was not reproduced

With the official code and defaults, HiSTaR reached a median ARI over the twelve DLPFC sections of 0.509 (mean 0.491, 95% bootstrap CI over sections 0.454–0.525) with the 3-start GMM, and 0.449 (mean, seed 0) when its embedding was clustered with R mclust as in the official pipeline, compared with a reported median of 0.65 over the twelve sections and 0.718 on section 151673 [6]. Fixing a bug that applies random masking at inference did not change accuracy (0.487; Supplementary Table S15). Possible causes include per-section tuning of the loss weights (described in the HiSTaR supplement), unpublished preprocessing and the choice of seed; the results above show that the clustering step alone can produce per-section differences of this size. [TODO: Report the outcome of contacting the authors.]

Released code of one method selects the reported epoch using the labels

[TODO: Contact the Spatial-MGCN authors and report their response before including this subsection.] Spatial-MGCN's released test scripts (`DLPFC_test.py`, `HBC_test.py`) compute ARI against the annotation at every training epoch and keep the best epoch for reporting. Applying the same selection rule in our runs (seed 0, 20 sections), this label-selected epoch had a mean ARI of 0.551 on DLPFC, 0.687 on STARmap and 0.221 on MERFISH, against 0.418, 0.303 and 0.182 for the final epoch used in our benchmark, an inflation of 0.147 on average (0.38 on STARmap; Supplementary Table S16). Selecting a run, epoch or seed by agreement with the labels is a form of label leakage that makes a method look competitive with the training-free baselines (0.551 against 0.558 on DLPFC) when its label-free result is not. An automated screen of the released repositories of 14 methods for other patterns of label use, with every candidate line read by hand, is reported in Supplementary Table S17 and Supplementary Note 5.

Discussion

We benchmarked 13 published methods and three training-free baselines on 20 sections from three technologies and separated the learned embedding from the clustering step. When all embeddings were clustered in the same way, standard neighbourhood smoothing had a higher mean ARI than most methods and could not be separated from the best of the rest (BASS, BANKSY, HiSTaR, SEDR and AnisoST); no method was shown to be equivalent to it. The clustering step was a larger source of variation than most differences between methods: random initialisation of the same mixture model spanned 0.18 ARI on a fixed embedding, the default fit of R mclust, used by several widely used methods, depended on a fixed but arbitrary random seed, and better likelihood did not reliably mean better domains. Reproducibility was limited both by the clustering step and by choices in released code, including selecting a reported epoch with the labels.

Why does simple smoothing do so well? On layered tissue such as cortex, domains are large, contiguous regions, and averaging each spot with its neighbours removes most of the sampling noise that separates spots of the same layer; much of what a graph encoder learns may be this smoothing. On single-cell MERFISH data, neighbouring cells often belong to different cell types, and the best-fitting mixture can then describe cell types rather than regions, which explains why likelihood and ARI disagree there. For a practitioner, the results suggest starting with a smoothing baseline, choosing the clustering step deliberately and reporting its variance. These results agree with recent benchmarks [13; 12; 11] and add a direct measurement of clustering variance on fixed embeddings for 14 methods across three technologies.

Recommendations.

(1) Report the clustering protocol: algorithm, implementation and version, initialisation, number of starts, random seeds and how they were chosen (Figure 4A,B). (2) Never select seeds, epochs or runs by agreement with the labels (Supplementary Table S16). (3) Report distributions over training seeds and over clustering runs or R seeds, with per-section values (Supplementary Tables S8–S10). (4) Compare encoders under the same clustering, and include training-free baselines such as neighbourhood smoothing (Figure 5). (5) Report run time, memory and seed variance alongside accuracy (Supplementary Table S13). (6) Release per-run results and seeds so that variance can be recomputed. (7) If fits are selected, use a label-free criterion and report how often it agrees with the annotation.

Limitations.

All methods were run with default or published settings, without a tuning budget, so the results do not show what each method can reach with its authors' own tuning (we could not reproduce HiSTaR's per-section tuning). The reference baseline was chosen after the means were seen, and the comparisons with it are exploratory. Four methods were run with one seed in the published pipelines and BayesSpace on Visium only (Supplementary Table S19), so seed variance is unavailable for them. The initialisation and R mclust experiments use one seed-0 embedding per section and method and do not include training-seed variance. The three datasets are brain tissues with laminar or regional structure; Slide-seq, Stereo-seq, Xenium and Visium HD data, larger sections and non-laminar tissues are not covered, and no simulated data were used. The STARmap and MERFISH annotations were distributed with the BASS analyses, and BASS was among the most accurate methods on these data; annotations may also have been derived with methods similar to the ones evaluated. The number of domains was given to every method. All runs used one CPU thread, so run times are not representative of GPU use. ARI is the primary metric; other metrics are in Supplementary Table S12.

Data availability

DLPFC: spatialLIBD (count matrices) and the SEDR analyses repository (layer annotations) [7]. STARmap and MERFISH with domain annotations: BASS-Analysis repository [18], from the original studies [16; 17]. Code, download scripts with checksums, per-run results and the scripts that generate every number, table and figure: [TODO: GitHub URL and Zenodo DOI].

Supplementary data

Supplementary Data are available at NAR Online.

Acknowledgements

Code, experiments and a first draft of this manuscript were produced with the assistance of an AI coding assistant (Claude, Anthropic); all results were checked by the authors, who take full responsibility for the content. [TODO: Confirm wording and add acknowledgements.]

Author contributions

[TODO: CRediT statement.]

Funding

[TODO: Funding, including funding for open access charge.]

Conflict of interest statement

The authors developed AnisoST, one of the training-free baselines. No other conflict of interest is declared.

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